

MSBD5001-Spring 2022

Predicting pneumonia in kidney transplant recipients

<https://www.kaggle.com/c/msbd5001-spring-2022/overview>

Description

Kidney transplantation is the optimal treatment to cure patients with end-stage renal disease (ESRD). However, infectious complication, especially pneumonia, is the main cause of mortality in the early stage. In this in-class competition, we aimed to study the association between collected patient immune status features during immune monitoring and pneumonia in kidney transplant patients through machine learning models.

The immune status features consist of the percentages and absolute cell counts of CD3+CD4+ T cells, CD3+CD8+ T cells, CD19+ B cells and natural killer (NK) cells, and median fluorescence intensity (MFI) of human leukocyte antigen (HLA)-DR on monocytes and CD64 on neutrophils. Also, basic information including age and sex is provided. The task is to predict whether the patient will get pneumonia after the kidney transplantation.

Dataset information

xxxxxx

ML problem definition

Multi-class classification

Evaluation Metric

The evaluation metric used is prediction accuracy.

Classifiers: (1) Decision Tree, (2) KNN and (3) Random Forest

Download data

```
In [ ]: # find the share link of the file/folder on Google Drive
file_share_links = [
    "https://drive.google.com/file/d/1pP81aU-10NWzVNBnNQUdVCeSrBL0kcWw/view?usp=sharing", #sample_submission.csv
    "https://drive.google.com/file/d/1UEZzWcKk_QaHwzDM16XdWGJLgkhiguj~view?usp=sharing", #test.csv
    "https://drive.google.com/file/d/1L7AYTx15-AQ2nXAuXm7hLC9HDvpKJCGw/view?usp=sharing", #train.csv
]

for file_share_link in file_share_links:
    # extract the ID of the file
    file_id = file_share_link[file_share_link.find("d/") + 2: file_share_link.find('/view')]
    print(file_id)

    # append the id to this REST command
    file_download_link = "https://docs.google.com/uc?export=download&id=" + file_id
    print(file_download_link)

1pP81aU-10NWzVNBnNQUdVCeSrBL0kcWw
https://docs.google.com/uc?export=download&id=1pP81aU-10NWzVNBnNQUdVCeSrBL0kcWw
1UEZzWcKk_QaHwzDM16XdWGJLgkhiguj~
https://docs.google.com/uc?export=download&id=1UEZzWcKk_QaHwzDM16XdWGJLgkhiguj~
1L7AYTx15-AQ2nXAuXm7hLC9HDvpKJCGw
https://docs.google.com/uc?export=download&id=1L7AYTx15-AQ2nXAuXm7hLC9HDvpKJCGw

In [ ]: !wget -O sample_submission.csv "https://docs.google.com/uc?export=download&id=1pP81aU-10NWzVNBnNQUdVCeSrBL0kcWw"
!wget -O test.csv "https://docs.google.com/uc?export=download&id=1UEZzWcKk_QaHwzDM16XdWGJLgkhiguj~"
!wget -O train.csv "https://docs.google.com/uc?export=download&id=1L7AYTx15-AQ2nXAuXm7hLC9HDvpKJCGw"

--2022-04-05 10:25:59-- https://docs.google.com/uc?export=download&id=1pP81aU-10NWzVNBnNQUdVCeSrBL0kcWw
Resolving docs.google.com (docs.google.com)... 108.177.13.100, 108.177.13.113, 108.177.13.102, ...
Connecting to docs.google.com (docs.google.com)|108.177.13.100|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://doc-0c-74-docs.googleusercontent.com/docs/securesc/ha0ro937gcuc7l7deffksulhg5h7mbp1/ns1s1s4kmid7aa2765bap8ph857gvbmc/1649154300000/1000462604393659472
9/*1pP81aU-10NWzVNBnNQUdVCeSrBL0kcWw?e=download [following]
Warning: wildcards not supported in HTTP.
--2022-04-05 10:26:00-- https://doc-0c-74-docs.googleusercontent.com/docs/securesc/ha0ro937gcuc7l7deffksulhg5h7mbp1/ns1s1s4kmid7aa2765bap8ph857gvbmc/1649154300000/10004
626043936594729/*1pP81aU-10NWzVNBnNQUdVCeSrBL0kcWw?e=download
Resolving doc-0c-74-docs.googleusercontent.com (doc-0c-74-docs.googleusercontent.com)... 173.194.215.132, 2607:f8b0:400c:c0c::84
Connecting to doc-0c-74-docs.googleusercontent.com (doc-0c-74-docs.googleusercontent.com)|173.194.215.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 294 [text/csv]
Saving to: 'sample_submission.csv'

sample_submission.c 100%[=====] 294 --.-KB/s in 0s

2022-04-05 10:26:00 (10.9 MB/s) - 'sample_submission.csv' saved [294/294]

--2022-04-05 10:26:00-- https://docs.google.com/uc?export=download&id=1UEZzWcKk_QaHwzDM16XdWGJLgkhiguj~
Resolving docs.google.com (docs.google.com)... 108.177.13.101, 108.177.13.138, 108.177.13.139, ...
Connecting to docs.google.com (docs.google.com)|108.177.13.101|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://doc-0k-74-docs.googleusercontent.com/docs/securesc/ha0ro937gcuc7l7deffksulhg5h7mbp1/ph07nf9rps7i1f53ji45v0vbfefo0hn73/1649154300000/1000462604393659472
9/*1UEZzWcKk_QaHwzDM16XdWGJLgkhiguj~?e=download [following]
Warning: wildcards not supported in HTTP.
--2022-04-05 10:26:01-- https://doc-0k-74-docs.googleusercontent.com/docs/securesc/ha0ro937gcuc7l7deffksulhg5h7mbp1/ph07nf9rps7i1f53ji45v0vbfefo0hn73/1649154300000/10004
626043936594729/*1UEZzWcKk_QaHwzDM16XdWGJLgkhiguj~?e=download
Resolving doc-0k-74-docs.googleusercontent.com (doc-0k-74-docs.googleusercontent.com)... 173.194.215.132, 2607:f8b0:400c:c0c::84
Connecting to doc-0k-74-docs.googleusercontent.com (doc-0k-74-docs.googleusercontent.com)|173.194.215.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4212 (4.1K) [text/csv]
Saving to: 'test.csv'

test.csv 100%[=====] 4.11K --.-KB/s in 0s

2022-04-05 10:26:00 (47.2 MB/s) - 'test.csv' saved [4212/4212]

--2022-04-05 10:26:01-- https://docs.google.com/uc?export=download&id=1L7AYTx15-AQ2nXAuXm7hLC9HDvpKJCGw
Resolving docs.google.com (docs.google.com)... 108.177.13.102, 108.177.13.139, 108.177.13.100, ...
Connecting to docs.google.com (docs.google.com)|108.177.13.102|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://doc-14-74-docs.googleusercontent.com/docs/securesc/ha0ro937gcuc7l7deffksulhg5h7mbp1/p66big71ja1s2s1rp3tkfh1j5e985k2n/1649154300000/1000462604393659472
9/*1L7AYTx15-AQ2nXAuXm7hLC9HDvpKJCGw?e=download [following]
Warning: wildcards not supported in HTTP.
--2022-04-05 10:26:01-- https://doc-14-74-docs.googleusercontent.com/docs/securesc/ha0ro937gcuc7l7deffksulhg5h7mbp1/p66big71ja1s2s1rp3tkfh1j5e985k2n/1649154300000/10004
626043936594729/*1L7AYTx15-AQ2nXAuXm7hLC9HDvpKJCGw?e=download
Resolving doc-14-74-docs.googleusercontent.com (doc-14-74-docs.googleusercontent.com)... 173.194.215.132, 2607:f8b0:400c:c0c::84
Connecting to doc-14-74-docs.googleusercontent.com (doc-14-74-docs.googleusercontent.com)|173.194.215.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 6299 (6.2K) [text/csv]
Saving to: 'train.csv'
```

```
train.csv          100%[=====] 6.15K --KB/s in 0s
2022-04-05 10:26:01 (39.6 MB/s) - 'train.csv' saved [6299/6299]
```

Setup

Install auto-ml pycaret lib

```
!pip install pycaret[full]
```

```
In [ ]: # !pip install pycaret
```

```
In [ ]: import os
import time
from typing import Iterable

import numpy as np
import pandas as pd
import seaborn as sb
import matplotlib.pyplot as plt

# save scikit-learn model
from joblib import dump, load
# import pickle

from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
from sklearn.metrics import precision_score, recall_score, f1_score, accuracy_score

#Split data
from sklearn.model_selection import train_test_split

from sklearn.preprocessing import MinMaxScaler
#Grid Search with Cross Validation
from sklearn.model_selection import GridSearchCV

# decision tree
from sklearn.tree import DecisionTreeClassifier

# knn
from sklearn.neighbors import KNeighborsClassifier

# random forest
from sklearn.ensemble import RandomForestClassifier
```

Data preparation

```
In [ ]: %sh
ls

head -n 5 train.csv
head -n 5 test.csv
head -n 5 sample_submission.csv
```

```
logs.log
sample_data
sample_submission.csv
test.csv
train.csv
id,MO_HLADR+ MFI (cells/ul),Neu CD64+MFI (cells/ul),CD3+T (cells/ul),CD8+T (cells/ul),CD4+T (cells/ul),NK (cells/ul),CD19+ (cells/ul),CD45+ (cells/ul),Age,Sex 0M1F,Mono
CD64+MFI (cells/ul),label
0,3556.0,2489.0,265.19,77.53,176.55,0.0,4.2,307.91,52.0,7515.0,1
1,1906.0,134.0,1442.61,551.9,876.07,112.1,168.15,1735.48,20,1,1756.0,0
2,1506.0,71.0,1332.74,684.2,655.26,244.95,216.52,1820.04,28,1,1311.0,0
3,683.0,94.0,419.23,255.8,162.17,72.05,44.68,538.22,55,1,1443.0,0
id,MO_HLADR+ MFI (cells/ul),Neu CD64+MFI (cells/ul),CD3+T (cells/ul),CD8+T (cells/ul),CD4+T (cells/ul),NK (cells/ul),CD19+ (cells/ul),CD45+ (cells/ul),Age,Sex 0M1F,Mono
CD64+MFI (cells/ul)
0,2843.0,156.0,1358.52,730.78,637.85,127.06,94.82,1588.62,45,1,3256.0
1,437.0,137.0,509.43,268.05,243.07,390.86,98.24,1002.76,51,1,491.0
2,826.0,82.0,1232.22,493.42,744.08,516.28,320.15,2200.58,32,0,1381.0
3,861.0,50.0,1512.86,925.51,590.07,380.25,25.8,1929.1,50,0,1377.0
id,label
0,0
1,0
2,0
3,0
```

```
In [ ]: folder_models = 'models'
folder_data = ''
path_train = folder_data + 'train.csv'
path_test = folder_data + 'test.csv'

#read train data
train = np.genfromtxt(path_train, delimiter=',', names=True, dtype=float)

row_index_name = 'id'
label_name = 'label'
feature_names = [x for x in train.dtype.names if (x != label_name) & (x != row_index_name)]

# train_x = train[feature_names].tolist()
# train_y = train[label_name].tolist()

#read test data
test = np.genfromtxt(path_test, delimiter=',', names=True, dtype=float)

# test_x = test[feature_names].tolist()
# test_y = test[label_name].tolist()

#class labels
labels = list(set(train[label_name]))
print(f'Classes/Labels of dataset (column: {label_name}):', labels)

# View
print(f'row_index_name: {row_index_name}')
print(f'label_name: {label_name}')
print(f'feature columns: {feature_names}')

# print(test_x[:5])
# print('Labels of test data:', test_y[:5])

print('train.shape =', train.shape)
print('test.shape =', test.shape)
```

```
Classes/Labels of dataset (column: label): [0.0, 1.0]
row_index_name: id
label_name: label
```

feature columns: ['MO_HLADR_MFI_cellsul', 'Neu_CD64MFI_cellsul', 'CD3T_cellsul', 'CD8T_cellsul', 'CD4T_cellsul', 'NK_cellsul', 'CD19_cellsul', 'CD45_cellsul', 'Age', 'Sex_0M1F', 'Mono_CD64MFI_cellsul']
x_0M1F', 'Mono_CD64MFI_cellsul']
train.shape = (87,)
test.shape = (59,)

In []:

```
# Convert np.array to df
df_train = pd.DataFrame(train)
df_test = pd.DataFrame(test)

if row_index_name:
    df_train.set_index(row_index_name, inplace=True)

    df_test.set_index(row_index_name, inplace=True)
    df_test.index = df_test.index.astype(int)

display(df_train.describe())
display(df_train.info())

display(df_test.describe())
display(df_test.info())
```

	MO_HLADR_MFI_cellsul	Neu_CD64MFI_cellsul	CD3T_cellsul	CD8T_cellsul	CD4T_cellsul	NK_cellsul	CD19_cellsul	CD45_cellsul	Age	Sex_0M1F	Mono_CD64MFI_cellsul	label
count	86.000000	86.000000	87.000000	87.000000	87.000000	87.000000	87.000000	87.000000	87.000000	87.000000	86.000000	87.000000
mean	1264.244186	290.383721	982.570115	479.340920	494.904023	212.732874	118.780920	1325.096437	40.218391	0.482759	2066.534884	0.333333
std	765.452376	490.283499	617.332545	344.326452	311.836604	173.553264	96.218344	791.602538	10.461919	0.502599	1198.401364	0.474137
min	112.000000	30.000000	74.400000	36.610000	39.590000	0.000000	4.200000	209.250000	19.000000	0.000000	72.000000	0.000000
25%	685.500000	77.500000	549.390000	237.920000	272.745000	78.815000	52.425000	780.615000	33.000000	0.000000	1461.250000	0.000000
50%	1108.500000	124.500000	871.710000	423.270000	459.720000	188.780000	89.790000	1179.270000	41.000000	0.000000	1757.500000	0.000000
75%	1602.250000	244.500000	1268.085000	624.450000	624.360000	262.845000	155.450000	1617.725000	49.500000	1.000000	2238.250000	1.000000
max	4145.000000	3124.000000	3791.230000	2548.100000	1517.810000	878.040000	485.860000	4757.280000	60.000000	1.000000	7515.000000	1.000000

```
<class 'pandas.core.frame.DataFrame'>
Float64Index: 87 entries, 0.0 to 86.0
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   MO_HLADR_MFI_cellsul    86 non-null     float64
1   Neu_CD64MFI_cellsul     86 non-null     float64
2   CD3T_cellsul            87 non-null     float64
3   CD8T_cellsul            87 non-null     float64
4   CD4T_cellsul            87 non-null     float64
5   NK_cellsul              87 non-null     float64
6   CD19_cellsul            87 non-null     float64
7   CD45_cellsul            87 non-null     float64
8   Age                     87 non-null     float64
9   Sex_0M1F                87 non-null     float64
10  Mono_CD64MFI_cellsul    86 non-null     float64
11  label                   87 non-null     float64
dtypes: float64(12)
memory usage: 8.8 KB
None
```

	MO_HLADR_MFI_cellsul	Neu_CD64MFI_cellsul	CD3T_cellsul	CD8T_cellsul	CD4T_cellsul	NK_cellsul	CD19_cellsul	CD45_cellsul	Age	Sex_0M1F	Mono_CD64MFI_cellsul	label
count	59.000000	59.000000	59.000000	59.000000	59.000000	59.000000	59.000000	59.000000	59.000000	59.000000	59.000000	59.000000
mean	1212.423729	206.491525	1085.340508	546.220339	523.237966	226.820339	115.048983	1439.65339	41.186441	0.355932	1971.220339	
std	772.139285	248.195027	564.337155	342.370020	271.730902	189.056327	87.200827	689.02181	9.438503	0.482905	1137.384129	
min	82.000000	24.000000	258.010000	114.980000	80.390000	17.720000	2.960000	314.25000	15.000000	0.000000	371.000000	
25%	696.500000	65.000000	629.890000	268.300000	336.955000	88.330000	59.500000	914.84000	34.500000	0.000000	1283.500000	
50%	1010.000000	114.000000	1025.320000	433.610000	511.000000	174.860000	98.240000	1378.32000	42.000000	0.000000	1701.000000	
75%	1623.000000	232.000000	1495.395000	751.380000	676.530000	318.140000	143.560000	1855.05000	49.000000	1.000000	2375.000000	
max	4195.000000	1141.000000	2771.200000	1738.550000	1225.680000	956.780000	501.910000	3355.86000	62.000000	1.000000	6788.000000	

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 59 entries, 0 to 58
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   MO_HLADR_MFI_cellsul    59 non-null     float64
1   Neu_CD64MFI_cellsul     59 non-null     float64
2   CD3T_cellsul            59 non-null     float64
3   CD8T_cellsul            59 non-null     float64
4   CD4T_cellsul            59 non-null     float64
5   NK_cellsul              59 non-null     float64
6   CD19_cellsul            59 non-null     float64
7   CD45_cellsul            59 non-null     float64
8   Age                     59 non-null     float64
9   Sex_0M1F                59 non-null     float64
10  Mono_CD64MFI_cellsul    59 non-null     float64
dtypes: float64(11)
memory usage: 5.5 KB
None
```

Clean data - Remove null rows

In []:

```
# Cheak `null` rows
display(df_train[df_train.isna().any(axis=1)])
display(df_train[df_train.isnull().any(axis=1)])

display(df_test[df_test.isna().any(axis=1)])
display(df_test[df_test.isnull().any(axis=1)])
```

MO_HLADR_MFI_cellsul	Neu_CD64MFI_cellsul	CD3T_cellsul	CD8T_cellsul	CD4T_cellsul	NK_cellsul	CD19_cellsul	CD45_cellsul	Age	Sex_0M1F	Mono_CD64MFI_cellsul	label
id											
39.0	NaN	NaN	1336.54	739.71	550.3	68.46	192.07	1615.68	21.0	0.0	NaN 0.0
MO_HLADR_MFI_cellsul Neu_CD64MFI_cellsul CD3T_cellsul CD8T_cellsul CD4T_cellsul NK_cellsul CD19_cellsul CD45_cellsul Age Sex_0M1F Mono_CD64MFI_cellsul label											
id											
39.0	NaN	NaN	1336.54	739.71	550.3	68.46	192.07	1615.68	21.0	0.0	NaN 0.0
MO_HLADR_MFI_cellsul Neu_CD64MFI_cellsul CD3T_cellsul CD8T_cellsul CD4T_cellsul NK_cellsul CD19_cellsul CD45_cellsul Age Sex_0M1F Mono_CD64MFI_cellsul											
id											
MO_HLADR_MFI_cellsul Neu_CD64MFI_cellsul CD3T_cellsul CD8T_cellsul CD4T_cellsul NK_cellsul CD19_cellsul CD45_cellsul Age Sex_0M1F Mono_CD64MFI_cellsul											
id											

In []:

```
# Remove `null` rows
## Train data
```

```
df_train = df_train[~df_train.isna().any(axis=1)]
display(df_train[df_train.isna().any(axis=1)])
```

```
## Test data
df_test = df_test[~df_test.isna().any(axis=1)]
display(df_test[df_test.isna().any(axis=1)])
```

```
MO_HLADR_MFI_cellul Neu_CD64MFI_cellul CD3T_cellul CD8T_cellul CD4T_cellul NK_cellul CD19_cellul CD45_cellul Age Sex_OM1F Mono_CD64MFI_cellul label
id

MO_HLADR_MFI_cellul Neu_CD64MFI_cellul CD3T_cellul CD8T_cellul CD4T_cellul NK_cellul CD19_cellul CD45_cellul Age Sex_OM1F Mono_CD64MFI_cellul
id
```

Split validation sets from train sets

randomly select rows from Pandas DataFrame,

- <https://www.geeksforgeeks.org/how-to-randomly-select-rows-from-pandas-dataframe/>

```
In [ ]: # To get 3 random rows
# each time it gives 3 different rows
# df_validation = df_train.sample(n = 3)

df_train, df_validation = train_test_split(df_train, test_size=0.08, random_state=200)

display(df_train.index)
display(df_validation.index)

Float64Index([ 9.0, 84.0, 19.0, 38.0, 2.0, 30.0, 10.0, 42.0, 61.0, 17.0, 51.0,
44.0, 5.0, 50.0, 29.0, 65.0, 64.0, 66.0, 68.0, 67.0, 34.0, 47.0,
25.0, 59.0, 36.0, 70.0, 54.0, 18.0, 41.0, 4.0, 12.0, 74.0, 79.0,
78.0, 86.0, 49.0, 8.0, 33.0, 48.0, 63.0, 3.0, 0.0, 21.0, 20.0,
82.0, 46.0, 32.0, 31.0, 53.0, 35.0, 76.0, 75.0, 71.0, 85.0, 13.0,
55.0, 22.0, 72.0, 24.0, 23.0, 73.0, 15.0, 27.0, 52.0, 7.0, 1.0,
57.0, 83.0, 6.0, 11.0, 58.0, 14.0, 80.0, 77.0, 56.0, 43.0, 69.0,
16.0, 26.0],
dtype='float64', name='id')
Float64Index([81.0, 28.0, 60.0, 37.0, 62.0, 40.0, 45.0], dtype='float64', name='id')
```

Make np.array (1) train_x train_y (2) validation_x validation_y

```
In [ ]: train_x = df_train[feature_names]
train_y = df_train[label_name]

test_x = validation_x = df_validation[feature_names]
test_y = validation_y = df_validation[label_name]

train_x
```

```
Out [ ]: MO_HLADR_MFI_cellul Neu_CD64MFI_cellul CD3T_cellul CD8T_cellul CD4T_cellul NK_cellul CD19_cellul CD45_cellul Age Sex_OM1F Mono_CD64MFI_cellul
id
9.0 1412.0 243.0 1177.19 684.42 490.50 185.30 67.22 1441.06 36.0 1.0 1213.0
84.0 634.0 1002.0 1300.00 558.00 724.00 67.00 105.00 1484.26 34.0 0.0 2926.0
19.0 403.0 555.0 313.48 131.53 182.69 46.68 7.90 370.30 40.0 0.0 2209.0
38.0 1010.0 1384.0 570.13 312.90 233.84 80.17 31.18 702.08 56.0 1.0 5501.0
2.0 1586.0 71.0 1332.74 684.20 655.26 244.95 216.52 1820.04 28.0 1.0 1311.0
... ... ... ... ... ... ... ... ... ... ...
56.0 1055.0 87.0 913.42 410.52 507.04 77.46 43.08 1040.18 48.0 0.0 1728.0
43.0 1679.0 79.0 483.21 162.00 309.00 227.05 101.09 817.24 39.0 0.0 4480.0
69.0 1495.0 125.0 2910.03 1431.78 1517.81 446.94 401.45 3817.75 20.0 1.0 1793.0
16.0 1045.0 124.0 1179.07 522.49 661.78 210.96 93.63 1498.11 49.0 1.0 2075.0
26.0 2088.0 98.0 962.30 533.03 459.72 370.88 142.89 1477.31 51.0 1.0 1599.0
```

79 rows x 11 columns

Comparison of Classifiers

```
In [ ]: # from pycaret.classification import *

# clf1 = setup(data = df_train,
#             target = label_name,
#             categorical_features = feature_names,
#             silent = True)
```

```
In [ ]: # compare_models(fold = 25) # 3 rows of training data are selected as validation set per each folding
```

```
In [ ]: # create_model('rf')
```

Re-install auto-ml pycaret lib

```
In [ ]: # 'pip install numpy
# 'pip install tf-estimator-nightly==2.8.0.dev2021122109
# 'pip install pycaret[full]
```

```
!pip install --use-deprecated=legacy-resolver pycaret[full]
```

```
Requirement already satisfied: pycaret[full] in /usr/local/lib/python3.7/dist-packages (2.3.9)
Requirement already satisfied: yellowbrick<1.0.1 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.3.post1)
Requirement already satisfied: joblib in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.0.1)
Requirement already satisfied: textblob in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.15.3)
Requirement already satisfied: kmodes>=0.10.1 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.12.0)
Requirement already satisfied: lightgbm>=2.3.1 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (3.3.2)
Requirement already satisfied: cufflinks>=0.17.0 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (5.4.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (3.2.2)
Requirement already satisfied: IPython in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (5.5.0)
Requirement already satisfied: Boruta in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.3)
Requirement already satisfied: cufflinks>=0.17.0 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.17.3)
Requirement already satisfied: pandas in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.3.5)
Requirement already satisfied: mlxtend>=0.17.0 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.19.0)
Requirement already satisfied: pyod in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.9.9)
Requirement already satisfied: pyLDAvis in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (3.2.2)
Requirement already satisfied: seaborn in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.11.2)
```

Requirement already satisfied: ipywidgets in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (7.7.0)
Requirement already satisfied: umap-learn in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.5.2)
Requirement already satisfied: numba<0.55 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.51.2)
Requirement already satisfied: pandas-profiling>=2.8.0 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (3.1.0)
Requirement already satisfied: scipy<=1.5.4 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.5.4)
Requirement already satisfied: nltk in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (3.2.5)
Requirement already satisfied: spacy<2.4.0 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (2.2.4)
Requirement already satisfied: mlflow in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.24.0)
Requirement already satisfied: scikit-learn==0.23.2 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.23.2)
Requirement already satisfied: imbalanced-learn==0.7.0 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.7.0)
Requirement already satisfied: gensim<4.0.0 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (3.6.0)
Requirement already satisfied: wordcloud in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.5.0)
Requirement already satisfied: plotly>=4.4.1 in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (5.5.0)
Requirement already satisfied: scikit-plot in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.3.7)
Requirement already satisfied: autoviz; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.1.37)
Requirement already satisfied: google-cloud-storage; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.18.1)
Requirement already satisfied: m2cgen; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.9.0)
Requirement already satisfied: fugue>=0.6.5; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.6.5)
Requirement already satisfied: fastapi; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.75.1)
Requirement already satisfied: hyperopt; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.1.2)
Requirement already satisfied: uvicorn; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.17.6)
Requirement already satisfied: catboost==0.23.2; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.0.4)
Requirement already satisfied: scikit-optimize>=0.8.1; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.9.0)
Requirement already satisfied: fairlearn; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.7.0)
Requirement already satisfied: ray[tune]>=1.0.0; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.11.0)
Requirement already satisfied: boto3; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.21.33)
Requirement already satisfied: optuna>=2.2.0; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (2.10.0)
Requirement already satisfied: azure-storage-blob; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (12.11.0)
Requirement already satisfied: explainerdashboard; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.3.8.2)
Requirement already satisfied: psutil; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (5.4.8)
Requirement already satisfied: xgboost>=1.1.0; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (1.5.2)
Requirement already satisfied: shap; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.40.0)
Requirement already satisfied: interpret>=0.2.4; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.2.4)
Requirement already satisfied: evidently; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.1.47.dev1)
Requirement already satisfied: tune-sklearn>=0.2.1; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (0.4.2)
Requirement already satisfied: gradio; extra == "full" in /usr/local/lib/python3.7/dist-packages (from pycaret[full]) (2.9.1)
Requirement already satisfied: cyclops>=0.10.0 in /usr/local/lib/python3.7/dist-packages (from yellowbrick>=1.0.1->pycaret[full]) (0.11.0)
Requirement already satisfied: numpty<1.20,>=1.16.0 in /usr/local/lib/python3.7/dist-packages (from yellowbrick>=1.0.1->pycaret[full]) (1.19.5)
Requirement already satisfied: wheel in /usr/local/lib/python3.7/dist-packages (from lightgbm>=2.3.1->pycaret[full]) (0.37.1)
Requirement already satisfied: python-dateutil>=2.1 in /usr/local/lib/python3.7/dist-packages (from matplotlib->pycaret[full]) (2.8.2)
Requirement already satisfied: pybars3>=2.0.4,!=2.1.2,!=2.1.6,>=2.0.1 in /usr/local/lib/python3.7/dist-packages (from matplotlib->pycaret[full]) (3.0.7)
Requirement already satisfied: kiwisolver>=1.0.1 in /usr/local/lib/python3.7/dist-packages (from matplotlib->pycaret[full]) (1.4.0)
Requirement already satisfied: pygments in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (2.6.1)
Requirement already satisfied: setuptools>=18.5 in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (57.4.0)
Requirement already satisfied: decorator in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (4.4.2)
Requirement already satisfied: simplegeneric>=0.8 in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (0.8.1)
Requirement already satisfied: pickleshare in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (0.7.5)
Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (5.1.1)
Requirement already satisfied: prompt-toolkit<2.0,>=1.0.4 in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (1.0.18)
Requirement already satisfied: pexpect; sys_platform != "win32" in /usr/local/lib/python3.7/dist-packages (from IPython->pycaret[full]) (4.8.0)
Requirement already satisfied: colorlover>=0.2.1 in /usr/local/lib/python3.7/dist-packages (from cufflinks>=0.17.0->pycaret[full]) (0.3.0)
Requirement already satisfied: six>=1.9.0 in /usr/local/lib/python3.7/dist-packages (from cufflinks>=0.17.0->pycaret[full]) (1.15.0)
Requirement already satisfied: pytz>=2017.3 in /usr/local/lib/python3.7/dist-packages (from pandas->pycaret[full]) (2018.9)
Requirement already satisfied: statsmodels in /usr/local/lib/python3.7/dist-packages (from pyod->pycaret[full]) (0.10.2)
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Requirement already satisfied: numexpr in /usr/local/lib/python3.7/dist-packages (from pyLDavis->pycaret[full]) (2.8.1)
Requirement already satisfied: future in /usr/local/lib/python3.7/dist-packages (from pyLDavis->pycaret[full]) (0.16.0)
Requirement already satisfied: jinjale>=2.7.2 in /usr/local/lib/python3.7/dist-packages (from pyLDavis->pycaret[full]) (2.11.3)
Requirement already satisfied: ipykernel>=4.5.1 in /usr/local/lib/python3.7/dist-packages (from ipywidgets->pycaret[full]) (4.10.1)
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Requirement already satisfied: umap-learn>=0.5 in /usr/local/lib/python3.7/dist-packages (from umap-learn->pycaret[full]) (0.5.6)
Requirement already satisfied: llvmlite>=0.35,>=0.34.0.dev0 in /usr/local/lib/python3.7/dist-packages (from numba<0.55->pycaret[full]) (0.34.0)
Requirement already satisfied: missingno>=0.4.2 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (0.5.1)
Requirement already satisfied: tangled-up-in-unicode==0.1.0 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (0.1.0)
Requirement already satisfied: phik>=0.11.1 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (0.12.2)
Requirement already satisfied: markupsafe>=2.0.1 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (2.0.1)
Requirement already satisfied: visions[type_image_path]==0.7.4 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (0.7.4)
Requirement already satisfied: htmlmin>=0.1.12 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (0.1.12)
Requirement already satisfied: requests>=2.24.0 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (2.27.1)
Requirement already satisfied: multimethod>=1.4 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (1.7)
Requirement already satisfied: pydantic>=1.8.1 in /usr/local/lib/python3.7/dist-packages (from pandas-profiling>=2.8.0->pycaret[full]) (1.9.0)
Requirement already satisfied: rsly<1.1.0,>=1.0.2 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (1.0.5)
Requirement already satisfied: plac<1.2.0,>=0.9.6 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (1.1.3)
Requirement already satisfied: thinc>=7.4.0 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (7.4.0)
Requirement already satisfied: wasabi<1.1.0,>=0.4.0 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (0.9.0)
Requirement already satisfied: cytoolz>=0.2.0 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (0.6.0)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (3.0.6)
Requirement already satisfied: blis<0.5.0,>=0.4.0 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (0.4.1)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (1.0.6)
Requirement already satisfied: catalogue<1.1.0,>=0.0.7 in /usr/local/lib/python3.7/dist-packages (from spacy<2.4.0->pycaret[full]) (1.0.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.7/dist-packages (from mlflow->pycaret[full]) (21.3)
Requirement already satisfied: databricks-cli>=0.8.7 in /usr/local/lib/python3.7/dist-packages (from mlflow->pycaret[full]) (0.16.6)
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Requirement already satisfied: importlib-metadata>=4.7.0,>=3.7.0 in /usr/local/lib/python3.7/dist-packages (from mlflow->pycaret[full]) (4.11.3)
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Requirement already satisfied: smart-open>=1.2.1 in /usr/local/lib/python3.7/dist-packages (from gensim<4.0.0->pycaret[full]) (5.2.1)
Requirement already satisfied: pillow in /usr/local/lib/python3.7/dist-packages (from wordcloud->pycaret[full]) (7.1.2)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.7/dist-packages (from plotly>=4.4.1->pycaret[full]) (8.0.1)
Requirement already satisfied: holoviews>=1.14.6 in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (1.14.6)
Requirement already satisfied: xlrd in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (1.1.0)
Requirement already satisfied: bokeh>=2.4.2 in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (2.4.2)
Requirement already satisfied: fsspec>=0.8.3 in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (0.8.3)
Requirement already satisfied: emoji in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (1.7.0)
Requirement already satisfied: panel>=0.12.6 in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (0.12.6)
Requirement already satisfied: jupyter in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (1.0.0)
Requirement already satisfied: hvplot>=0.7.3 in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (0.7.3)
Requirement already satisfied: typing-extensions>=4.1.1 in /usr/local/lib/python3.7/dist-packages (from autoviz; extra == "full"->pycaret[full]) (4.1.1)
Requirement already satisfied: google-resumable-media<0.5.0dev,>=0.3.1 in /usr/local/lib/python3.7/dist-packages (from google-cloud-storage; extra == "full"->pycaret[full]) (0.4.1)
Requirement already satisfied: google-cloud-core<2.0dev,>=1.0.0 in /usr/local/lib/python3.7/dist-packages (from google-cloud-storage; extra == "full"->pycaret[full]) (1.0.3)
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Requirement already satisfied: qpd>=0.2.6 in /usr/local/lib/python3.7/dist-packages (from fugue>=0.6.5; extra == "full"->pycaret[full]) (0.2.6)
Requirement already satisfied: pyarrow>=0.15.1 in /usr/local/lib/python3.7/dist-packages (from fugue>=0.6.5; extra == "full"->pycaret[full]) (6.0.1)
Requirement already satisfied: triad>=0.6.0 in /usr/local/lib/python3.7/dist-packages (from fugue>=0.6.5; extra == "full"->pycaret[full]) (0.6.0)
Requirement already satisfied: starlette>=0.17.1 in /usr/local/lib/python3.7/dist-packages (from fastapi; extra == "full"->pycaret[full]) (0.17.1)
Requirement already satisfied: networkx in /usr/local/lib/python3.7/dist-packages (from hyperopt; extra == "full"->pycaret[full]) (2.6.3)
Requirement already satisfied: pymongo in /usr/local/lib/python3.7/dist-packages (from hyperopt; extra == "full"->pycaret[full]) (4.0.2)
Requirement already satisfied: wasabi>=0.4.0 in /usr/local/lib/python3.7/dist-packages (from uvicorn; extra == "full"->pycaret[full]) (3.5.0)
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.7/dist-packages (from uvicorn; extra == "full"->pycaret[full]) (0.13.0)
Requirement already satisfied: graphviz in /usr/local/lib/python3.7/dist-packages (from catboost>=0.23.2; extra == "full"->pycaret[full]) (0.10.1)
Requirement already satisfied: pyaml>=16.9 in /usr/local/lib/python3.7/dist-packages (from scikit-optimize>=0.8.1; extra == "full"->pycaret[full]) (21.10.1)
Requirement already satisfied: msgpack<2.0.0,>=1.0.0 in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (1.0.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (3.6.0)
Requirement already satisfied: redis>=3.5.0 in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (4.2.2)
Requirement already satisfied: attrs in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (21.4.0)
Requirement already satisfied: jsonschema in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (4.3.3)
Requirement already satisfied: pyprio<=1.43.0,>=1.28.1 in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (1.43.0)
Requirement already satisfied: tensorflow in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (2.5)
Requirement already satisfied: tabulate; extra == "tune" in /usr/local/lib/python3.7/dist-packages (from ray[tune]>=1.0.0; extra == "full"->pycaret[full]) (0.8.9)
Requirement already satisfied: botocore<1.25.0,>=1.24.33 in /usr/local/lib/python3.7/dist-packages (from boto3; extra == "full"->pycaret[full]) (1.24.33)
Requirement already satisfied: s3transfer<0.6.0,>=0.5.0 in /usr/local/lib/python3.7/dist-packages (from boto3; extra == "full"->pycaret[full]) (0.5.2)

Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/local/lib/python3.7/dist-packages (from boto3; extra == "full"-->pycaret[full]) (1.0.0)

Requirement already satisfied: cliff in /usr/local/lib/python3.7/dist-packages (from optuna==2.0.0; extra == "full"-->pycaret[full]) (3.10.1)

Requirement already satisfied: cmaes<0.8.2 in /usr/local/lib/python3.7/dist-packages (from optuna==2.0.0; extra == "full"-->pycaret[full]) (0.8.2)

Requirement already satisfied: colorlog in /usr/local/lib/python3.7/dist-packages (from optuna==2.0.0; extra == "full"-->pycaret[full]) (6.6.0)

Requirement already satisfied: cryptography>=2.1.4 in /usr/local/lib/python3.7/dist-packages (from azure-storage-blob; extra == "full"-->pycaret[full]) (36.0.2)

Requirement already satisfied: msrest<0.6.21 in /usr/local/lib/python3.7/dist-packages (from azure-storage-blob; extra == "full"-->pycaret[full]) (0.6.21)

Requirement already satisfied: azure-core<2.0.0,>=1.15.0 in /usr/local/lib/python3.7/dist-packages (from azure-storage-blob; extra == "full"-->pycaret[full]) (1.23.1)

Requirement already satisfied: dash-bootstrap-components<1 in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (0.13.1)

Requirement already satisfied: flask-simplelogin in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (0.1.1)

Requirement already satisfied: dash-auth in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (1.4.1)

Requirement already satisfied: waitress in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (2.1.1)

Requirement already satisfied: Werkzeug<2.0.3 in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (1.0.1)

Requirement already satisfied: oyaml in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (1.0)

Requirement already satisfied: dtreeviz<1.3 in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (1.3.5)

Requirement already satisfied: dash==2 in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (2.3.1)

Requirement already satisfied: jupyter-dash in /usr/local/lib/python3.7/dist-packages (from explainerdashboard; extra == "full"-->pycaret[full]) (0.4.2)

Requirement already satisfied: slicer<0.0.7 in /usr/local/lib/python3.7/dist-packages (from shap; extra == "full"-->pycaret[full]) (0.0.7)

Requirement already satisfied: interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4 in /usr/local/lib/python3.7/dist-packages (from interpret==0.2.4; extra == "full"-->pycaret[full]) (0.2.7)

Requirement already satisfied: dataclasses<=0.6 in /usr/local/lib/python3.7/dist-packages (from evidently; extra == "full"-->pycaret[full]) (0.6)

Requirement already satisfied: orjson in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (3.6.7)

Requirement already satisfied: ffmpeg in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (0.3.0)

Requirement already satisfied: python-multipart in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (0.0.5)

Requirement already satisfied: pycryptodome in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (3.14.1)

Requirement already satisfied: pydub in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (0.25.1)

Requirement already satisfied: markdown-it-py[linkify,plugins] in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (2.0.1)

Requirement already satisfied: analytics-python in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (1.4.0)

Requirement already satisfied: paramiko in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (2.10.3)

Requirement already satisfied: aiohttp in /usr/local/lib/python3.7/dist-packages (from gradio; extra == "full"-->pycaret[full]) (3.8.1)

Requirement already satisfied: cwidth in /usr/local/lib/python3.7/dist-packages (from prompt-toolkit<2.0.0,>=1.0.4->IPython->pycaret[full]) (0.2.5)

Requirement already satisfied: ptyprocess<0.5 in /usr/local/lib/python3.7/dist-packages (from pexpect; sys_platform != "win32"-->IPython->pycaret[full]) (0.7.0)

Requirement already satisfied: patsy<0.4.0 in /usr/local/lib/python3.7/dist-packages (from statsmodels->pyod->pycaret[full]) (0.5.2)

Requirement already satisfied: jupyter-client in /usr/local/lib/python3.7/dist-packages (from ipykernel==4.5.1->ipywidgets->pycaret[full]) (5.3.5)

Requirement already satisfied: tornado<4.0 in /usr/local/lib/python3.7/dist-packages (from ipykernel==4.5.1->ipywidgets->pycaret[full]) (5.1.1)

Requirement already satisfied: jupyter-core in /usr/local/lib/python3.7/dist-packages (from nbformat>4.2.0->ipywidgets->pycaret[full]) (4.9.2)

Requirement already satisfied: notebook<4.4.1 in /usr/local/lib/python3.7/dist-packages (from widgetsnextension<3.6.0->ipywidgets->pycaret[full]) (5.3.1)

Requirement already satisfied: imagehash; extra == "type_image_path" in /usr/local/lib/python3.7/dist-packages (from visions[type_image_path]=0.7.4->pandas-profiling==2.8.0->pycaret[full]) (4.2.1)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.7/dist-packages (from requests>=2.24.0->pandas-profiling==2.8.0->pycaret[full]) (2021.10.8)

Requirement already satisfied: charset-normalizer<=2.0.0; python_version >= "3" in /usr/local/lib/python3.7/dist-packages (from requests>=2.24.0->pandas-profiling==2.8.0->pycaret[full]) (2.0.12)

Requirement already satisfied: urllib3<1.27,>=1.21.1 in /usr/local/lib/python3.7/dist-packages (from requests>=2.24.0->pandas-profiling==2.8.0->pycaret[full]) (1.24.3)

Requirement already satisfied: idna<4,>=2.5; python_version >= "3" in /usr/local/lib/python3.7/dist-packages (from requests>=2.24.0->pandas-profiling==2.8.0->pycaret[full]) (2.10)

Requirement already satisfied: oauthlib<3.1.0 in /usr/local/lib/python3.7/dist-packages (from databricks-cli>=0.8.7->mlflow->pycaret[full]) (3.2.0)

Requirement already satisfied: pyjwt<1.7.0 in /usr/local/lib/python3.7/dist-packages (from databricks-cli>=0.8.7->mlflow->pycaret[full]) (2.3.0)

Requirement already satisfied: prometheus-client in /usr/local/lib/python3.7/dist-packages (from prometheus-flask-exporter->mlflow->pycaret[full]) (0.13.1)

Requirement already satisfied: greenlet<0.4.17; python_version >= "3" and platform_machine == "aarch64" or (platform_machine == "ppc64le" or (platform_machine == "x86_64" or (platform_machine == "amd64" or (platform_machine == "AMD64" or (platform_machine == "win32" or (platform_machine == "WIN32"))))) in /usr/local/lib/python3.7/dist-packages (from sqlalchemy==mlflow->pycaret[full]) (1.1.2)

Requirement already satisfied: importlib-resources; python_version < "3.9" in /usr/local/lib/python3.7/dist-packages (from alembic==mlflow->pycaret[full]) (5.4.0)

Requirement already satisfied: Mako in /usr/local/lib/python3.7/dist-packages (from alembic==mlflow->pycaret[full]) (1.2.0)

Requirement already satisfied: itsdangerous<2.0,>=0.24 in /usr/local/lib/python3.7/dist-packages (from Flask==mlflow->pycaret[full]) (1.1.0)

Requirement already satisfied: gitdb<5,>=4.0.1 in /usr/local/lib/python3.7/dist-packages (from gitpython>=2.1.0->mlflow->pycaret[full]) (4.0.9)

Requirement already satisfied: websocket-client==0.32.0 in /usr/local/lib/python3.7/dist-packages (from docker==4.0.0->mlflow->pycaret[full]) (1.3.2)

Requirement already satisfied: zipp<0.5 in /usr/local/lib/python3.7/dist-packages (from importlib-metadata==4.7.0,>=3.7.0->mlflow->pycaret[full]) (3.7.0)

Requirement already satisfied: colorcet in /usr/local/lib/python3.7/dist-packages (from holoviews==1.14.6->autoviz; extra == "full"-->pycaret[full]) (3.0.0)

Requirement already satisfied: param<2.0,>=1.9.3 in /usr/local/lib/python3.7/dist-packages (from holoviews==1.14.6->autoviz; extra == "full"-->pycaret[full]) (1.12.0)

Requirement already satisfied: pyviz-comms<0.7.4 in /usr/local/lib/python3.7/dist-packages (from holoviews==1.14.6->autoviz; extra == "full"-->pycaret[full]) (2.1.0)

Requirement already satisfied: markdown in /usr/local/lib/python3.7/dist-packages (from panel==0.12.6->autoviz; extra == "full"-->pycaret[full]) (3.3.6)

Requirement already satisfied: pycpt<0.4.4 in /usr/local/lib/python3.7/dist-packages (from panel==0.12.6->autoviz; extra == "full"-->pycaret[full]) (0.4.8)

Requirement already satisfied: bleach in /usr/local/lib/python3.7/dist-packages (from panel==0.12.6->autoviz; extra == "full"-->pycaret[full]) (4.1.0)

Requirement already satisfied: jupyter-console in /usr/local/lib/python3.7/dist-packages (from jupyter-autoviz; extra == "full"-->pycaret[full]) (5.2.0)

Requirement already satisfied: nbconvert in /usr/local/lib/python3.7/dist-packages (from jupyter-autoviz; extra == "full"-->pycaret[full]) (5.6.1)

Requirement already satisfied: qtconsole in /usr/local/lib/python3.7/dist-packages (from jupyter-autoviz; extra == "full"-->pycaret[full]) (5.2.2)

Requirement already satisfied: google-api-core<2.0.0dev,>=1.14.0 in /usr/local/lib/python3.7/dist-packages (from google-cloud-core<2.0dev,>=1.0.0->google-cloud-storage; extra == "full"-->pycaret[full]) (1.26.3)

Requirement already satisfied: cachetools<5.0,>=2.0.0 in /usr/local/lib/python3.7/dist-packages (from google-auth==1.2.0->google-cloud-storage; extra == "full"-->pycaret[full]) (4.2.4)

Requirement already satisfied: rsa<5,>=3.1.4; python_version >= "3.6" in /usr/local/lib/python3.7/dist-packages (from google-auth==1.2.0->google-cloud-storage; extra == "full"-->pycaret[full]) (4.8)

Requirement already satisfied: pyasn1-modules==0.2.1 in /usr/local/lib/python3.7/dist-packages (from google-auth==1.2.0->google-cloud-storage; extra == "full"-->pycaret[full]) (0.2.8)

Requirement already satisfied: antlr4-python3-runtime in /usr/local/lib/python3.7/dist-packages (from qpd>=0.2.6->fugue>=0.6.5; extra == "full"-->pycaret[full]) (4.9.3)

Requirement already satisfied: fs in /usr/local/lib/python3.7/dist-packages (from triad>=0.6.0->fugue>=0.6.5; extra == "full"-->pycaret[full]) (2.4.15)

Requirement already satisfied: ciso8601; platform_system != "Windows" in /usr/local/lib/python3.7/dist-packages (from triad>=0.6.0->fugue>=0.6.5; extra == "full"-->pycaret[full]) (2.2.0)

Requirement already satisfied: anyio<4,>=3.0.0 in /usr/local/lib/python3.7/dist-packages (from starlette==0.17.1->fastapi; extra == "full"-->pycaret[full]) (3.5.0)

Requirement already satisfied: async-timeout<4.0.2 in /usr/local/lib/python3.7/dist-packages (from redis>=3.5.0->ray[tune]>=1.0.0; extra == "full"-->pycaret[full]) (4.0.2)

Requirement already satisfied: deprecated<1.2.3 in /usr/local/lib/python3.7/dist-packages (from redis>=3.5.0->ray[tune]>=1.0.0; extra == "full"-->pycaret[full]) (1.2.13)

Requirement already satisfied: pyrsistent!=0.17.0,!=0.17.1,!=0.17.2,>=0.14.0 in /usr/local/lib/python3.7/dist-packages (from jsonschema->ray[tune]>=1.0.0; extra == "full"-->pycaret[full]) (0.18.1)

Requirement already satisfied: autopage<0.4.0 in /usr/local/lib/python3.7/dist-packages (from cliff--optuna==2.2.0; extra == "full"-->pycaret[full]) (0.5.0)

Requirement already satisfied: PrettyTable<0.7.2 in /usr/local/lib/python3.7/dist-packages (from cliff--optuna==2.2.0; extra == "full"-->pycaret[full]) (3.2.0)

Requirement already satisfied: pbr<2.1.0,>=2.0.0 in /usr/local/lib/python3.7/dist-packages (from cliff--optuna==2.2.0; extra == "full"-->pycaret[full]) (5.8.1)

Requirement already satisfied: cmd2<1.0.0 in /usr/local/lib/python3.7/dist-packages (from cliff--optuna==2.2.0; extra == "full"-->pycaret[full]) (2.4.0)

Requirement already satisfied: stevedore<2.0.1 in /usr/local/lib/python3.7/dist-packages (from cliff--optuna==2.2.0; extra == "full"-->pycaret[full]) (3.5.0)

Requirement already satisfied: cffi>=1.12 in /usr/local/lib/python3.7/dist-packages (from cryptography>=2.1.4->azure-storage-blob; extra == "full"-->pycaret[full]) (1.15.0)

Requirement already satisfied: requests-oauthlib>=0.5.0 in /usr/local/lib/python3.7/dist-packages (from msrest>=0.6.21->azure-storage-blob; extra == "full"-->pycaret[full]) (1.3.1)

Requirement already satisfied: isodate>=0.6.0 in /usr/local/lib/python3.7/dist-packages (from msrest>=0.6.21->azure-storage-blob; extra == "full"-->pycaret[full]) (0.6.1)

Requirement already satisfied: WTForms>=2.1 in /usr/local/lib/python3.7/dist-packages (from flask-simplelogin->explainerdashboard; extra == "full"-->pycaret[full]) (3.0.1)

Requirement already satisfied: Flask-WTF<0.16.0,>=0.15.1 in /usr/local/lib/python3.7/dist-packages (from flask-simplelogin->explainerdashboard; extra == "full"-->pycaret[full]) (0.15.1)

Requirement already satisfied: retrying in /usr/local/lib/python3.7/dist-packages (from dash-auth->explainerdashboard; extra == "full"-->pycaret[full]) (1.3.3)

Requirement already satisfied: chart-studio<1.0.0 in /usr/local/lib/python3.7/dist-packages (from dash-auth->explainerdashboard; extra == "full"-->pycaret[full]) (1.1.0)

Requirement already satisfied: flask-seasurf in /usr/local/lib/python3.7/dist-packages (from dash-auth->explainerdashboard; extra == "full"-->pycaret[full]) (1.1.1)

Requirement already satisfied: flask-compress in /usr/local/lib/python3.7/dist-packages (from dash-auth->explainerdashboard; extra == "full"-->pycaret[full]) (1.11)

Requirement already satisfied: ua-parser in /usr/local/lib/python3.7/dist-packages (from dash-auth->explainerdashboard; extra == "full"-->pycaret[full]) (0.10.0)

Requirement already satisfied: pytest in /usr/local/lib/python3.7/dist-packages (from dtreeviz>=1.3->explainerdashboard; extra == "full"-->pycaret[full]) (3.6.4)

Requirement already satisfied: colour in /usr/local/lib/python3.7/dist-packages (from dtreeviz>=1.3->explainerdashboard; extra == "full"-->pycaret[full]) (0.1.5)

Requirement already satisfied: dash-table<5.0.0 in /usr/local/lib/python3.7/dist-packages (from dash==2->explainerdashboard; extra == "full"-->pycaret[full]) (5.0.0)

Requirement already satisfied: dash-core-components==2.0.0 in /usr/local/lib/python3.7/dist-packages (from dash==2->explainerdashboard; extra == "full"-->pycaret[full]) (2.0.0)

Requirement already satisfied: dash-html-components==2.0.0 in /usr/local/lib/python3.7/dist-packages (from dash==2->explainerdashboard; extra == "full"-->pycaret[full]) (2.0.0)

Requirement already satisfied: nest-asyncio in /usr/local/lib/python3.7/dist-packages (from jupyter-dash->explainerdashboard; extra == "full"-->pycaret[full]) (1.5.4)

Requirement already satisfied: ansihtml in /usr/local/lib/python3.7/dist-packages (from jupyter-dash->explainerdashboard; extra == "full"-->pycaret[full]) (1.7.0)

Requirement already satisfied: event==1.3.6; extra == "dash" in /usr/local/lib/python3.7/dist-packages (from interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret==0.2.4; extra == "full"-->pycaret[full]) (21.12.0)

Requirement already satisfied: dash-cytoscape>=0.1.1; extra == "dash" in /usr/local/lib/python3.7/dist-packages (from interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret==0.2.4; extra == "full"-->pycaret[full]) (0.3.0)

Requirement already satisfied: lime>=0.1.1.33; extra == "lime" in /usr/local/lib/python3.7/dist-packages (from interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret==0.2.4; extra == "full"-->pycaret[full]) (0.2.0.1)

Requirement already satisfied: SALib<1.3.3; extra == "sensitivity" in /usr/local/lib/python3.7/dist-packages (from interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret==0.2.4; extra == "full"-->pycaret[full]) (1.4.5)

Requirement already satisfied: dill<=0.2.5; extra == "shap" in /usr/local/lib/python3.7/dist-packages (from interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret==0.2.4; extra == "full"-->pycaret[full]) (0.3.4)

Requirement already satisfied: skope-rules<=1.0.1; extra == "skoperules" in /usr/local/lib/python3.7/dist-packages (from interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret==0.2.4; extra == "full"-->pycaret[full]) (1.0.1)

Requirement already satisfied: treeinterpreter>=0.2.2; extra == "treeinterpreter" in /usr/local/lib/python3.7/dist-packages (from interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret==0.2.4; extra == "full"-->pycaret[full]) (0.2.3)

Requirement already satisfied: mdurl<=0.1 in /usr/local/lib/python3.7/dist-packages (from markdown-it-py[linkify,plugins]->gradio; extra == "full"-->pycaret[full]) (0.1.0)

Requirement already satisfied: linkify-it-py<=1.0; extra == "linkify" in /usr/local/lib/python3.7/dist-packages (from markdown-it-py[linkify,plugins]->gradio; extra == "full"-->pycaret[full]) (1.0.3)

Requirement already satisfied: mdit-py-plugins; extra == "plugins" in /usr/local/lib/python3.7/dist-packages (from markdown-it-py[linkify,plugins]->gradio; extra == "full"-->pycaret[full]) (0.3.0)

Requirement already satisfied: backoff<=1.10.0 in /usr/local/lib/python3.7/dist-packages (from analytics-python->gradio; extra == "full"-->pycaret[full]) (1.10.0)

Requirement already satisfied: monotonic<1.5 in /usr/local/lib/python3.7/dist-packages (from analytics-python->gradio; extra == "full"-->pycaret[full]) (1.6)

Requirement already satisfied: bcrypt>=3.1.3 in /usr/local/lib/python3.7/dist-packages (from paramiko->gradio; extra == "full"-->pycaret[full]) (3.2.0)

Requirement already satisfied: pynacl<1.0.1 in /usr/local/lib/python3.7/dist-packages (from paramiko->gradio; extra == "full"-->pycaret[full]) (1.5.0)

Requirement already satisfied: frozenlist<=1.1.1 in /usr/local/lib/python3.7/dist-packages (from aiohttp->gradio; extra == "full"-->pycaret[full]) (1.3.0)

Requirement already satisfied: asyncctest<=0.13.0; python_version < "3.8" in /usr/local/lib/python3.7/dist-packages (from aiohttp->gradio; extra == "full"-->pycaret[full]) (0.13.0)

Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.7/dist-packages (from aiohttp->gradio; extra == "full"-->pycaret[full]) (1.2.0)

Requirement already satisfied: yarl<2.0,>=1.0 in /usr/local/lib/python3.7/dist-packages (from aiohttp->gradio; extra == "full"->pycaret[full]) (1.7.2)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.7/dist-packages (from aiohttp->gradio; extra == "full"->pycaret[full]) (6.0.2)
Requirement already satisfied: pyzmq<13 in /usr/local/lib/python3.7/dist-packages (from jupyter-client->ipykernel<=4.5.1->ipywidgets->pycaret[full]) (22.3.0)
Requirement already satisfied: Send2Trash in /usr/local/lib/python3.7/dist-packages (from notebook<=4.4.1->widgetsnbextension<=3.6.0->ipywidgets->pycaret[full]) (1.8.0)
Requirement already satisfied: terminado<=0.8.1 in /usr/local/lib/python3.7/dist-packages (from notebook<=4.4.1->widgetsnbextension<=3.6.0->ipywidgets->pycaret[full]) (0.13.3)
Requirement already satisfied: PyWavelets in /usr/local/lib/python3.7/dist-packages (from imagehash; extra == "type_image_path"->visions[type_image_path]==0.7.4->pandas-profiling<=2.8.0->pycaret[full]) (1.3.0)
Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.7/dist-packages (from gitdb<5,>=4.0.1->gitpython<=2.1.0->mlflow->pycaret[full]) (5.0.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.7/dist-packages (from bleach->panel==0.12.6->autoviz; extra == "full"->pycaret[full]) (0.5.1)
Requirement already satisfied: defusedxml in /usr/local/lib/python3.7/dist-packages (from nbconvert->jupyter->autoviz; extra == "full"->pycaret[full]) (0.7.1)
Requirement already satisfied: mistune<2,>=0.8.1 in /usr/local/lib/python3.7/dist-packages (from nbconvert->jupyter->autoviz; extra == "full"->pycaret[full]) (0.8.4)
Requirement already satisfied: testpath in /usr/local/lib/python3.7/dist-packages (from nbconvert->jupyter->autoviz; extra == "full"->pycaret[full]) (0.6.0)
Requirement already satisfied: pandocfilters<=1.4.1 in /usr/local/lib/python3.7/dist-packages (from nbconvert->jupyter->autoviz; extra == "full"->pycaret[full]) (1.5.0)
Requirement already satisfied: qtpy in /usr/local/lib/python3.7/dist-packages (from qtconsole->jupyter->autoviz; extra == "full"->pycaret[full]) (2.0.1)
Requirement already satisfied: googleapis-common-protos<2.0dev,>=1.6.0 in /usr/local/lib/python3.7/dist-packages (from google-api-core<2.0.0dev,>=1.14.0->google-cloud-core<2.0dev,>=1.0.0->google-cloud-storage; extra == "full"->pycaret[full]) (1.56.0)
Requirement already satisfied: pyasn1<=0.1.3 in /usr/local/lib/python3.7/dist-packages (from google-api-core<2.0.0dev,>=1.14.0->google-cloud-storage; extra == "full"->pycaret[full]) (0.4.8)
Requirement already satisfied: appdirs<=1.4.3 in /usr/local/lib/python3.7/dist-packages (from fs->triad<=0.6.0->fugue<=0.6.5; extra == "full"->pycaret[full]) (1.4.4)
Requirement already satisfied: sniffio<=1.1 in /usr/local/lib/python3.7/dist-packages (from anyio<4,>=3.0.0->starlette==0.17.1->fastapi; extra == "full"->pycaret[full]) (1.2.0)
Requirement already satisfied: wrapt<2,>=1.10 in /usr/local/lib/python3.7/dist-packages (from deprecated<=1.2.3->redis<=3.5.0->ray[tune]<=1.0.0; extra == "full"->pycaret[full]) (1.14.0)
Requirement already satisfied: pyperclip<=1.6 in /usr/local/lib/python3.7/dist-packages (from cmd2<=1.0.0->cliff->optuna<=2.2.0; extra == "full"->pycaret[full]) (1.8.2)
Requirement already satisfied: pycparser in /usr/local/lib/python3.7/dist-packages (from cffi<=1.12->cryptography<=2.1.4->azure-storage-blob; extra == "full"->pycaret[full]) (2.21)
Requirement already satisfied: brotli in /usr/local/lib/python3.7/dist-packages (from flask-compress->dash-auth->explainerdashboard; extra == "full"->pycaret[full]) (1.0.9)
Requirement already satisfied: more-itertools<=4.0.0 in /usr/local/lib/python3.7/dist-packages (from pytest->dtreeviz<=1.3->explainerdashboard; extra == "full"->pycaret[full]) (8.12.0)
Requirement already satisfied: pluggy<0.8,>=0.5 in /usr/local/lib/python3.7/dist-packages (from pytest->dtreeviz<=1.3->explainerdashboard; extra == "full"->pycaret[full]) (0.7.1)
Requirement already satisfied: atomicwrites<=1.0 in /usr/local/lib/python3.7/dist-packages (from pytest->dtreeviz<=1.3->explainerdashboard; extra == "full"->pycaret[full]) (1.4.0)
Requirement already satisfied: py<=1.5.0 in /usr/local/lib/python3.7/dist-packages (from pytest->dtreeviz<=1.3->explainerdashboard; extra == "full"->pycaret[full]) (1.1.0)
Requirement already satisfied: zope.event in /usr/local/lib/python3.7/dist-packages (from gevent<=1.3.6; extra == "dash"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (4.5.0)
Requirement already satisfied: zope.interface in /usr/local/lib/python3.7/dist-packages (from gevent<=1.3.6; extra == "dash"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (5.4.0)
Requirement already satisfied: scikit-image<=0.12 in /usr/local/lib/python3.7/dist-packages (from lime<=0.1.1.33; extra == "lime"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (0.18.3)
Requirement already satisfied: pathos in /usr/local/lib/python3.7/dist-packages (from SALib<=1.3.3; extra == "sensitivity"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (0.2.8)
Requirement already satisfied: uc-micro-py in /usr/local/lib/python3.7/dist-packages (from linkify-it-py<=1.0; extra == "linkify"->markdown-it-py[linkify,plugins]->gradio; extra == "full"->pycaret[full]) (1.0.1)
Requirement already satisfied: imageio<=2.3.0 in /usr/local/lib/python3.7/dist-packages (from scikit-image<=0.12->lime<=0.1.1.33; extra == "lime"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (2.4.1)
Requirement already satisfied: tiffle<=2019.7.26 in /usr/local/lib/python3.7/dist-packages (from scikit-image<=0.12->lime<=0.1.1.33; extra == "lime"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (20.21.11.2)
Requirement already satisfied: pox<=0.3.0 in /usr/local/lib/python3.7/dist-packages (from pathos->SALib<=1.3.3; extra == "sensitivity"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (0.3.0)
Requirement already satisfied: multiprocessing<=0.70.12 in /usr/local/lib/python3.7/dist-packages (from pathos->SALib<=1.3.3; extra == "sensitivity"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (0.70.1.2.2)
Requirement already satisfied: ppft<=1.6.6.4 in /usr/local/lib/python3.7/dist-packages (from pathos->SALib<=1.3.3; extra == "sensitivity"->interpret-core[dash,debug,decisiontree,ebm,lime,linear,notebook,plotly,required,sensitivity,shap,skoperules,treeinterpreter]>=0.2.4->interpret<=0.2.4; extra == "full"->pycaret[full]) (1.6.6.4)

In [1]:

```
from pycaret.classification import *

clf1 = setup(data = df_train,
             target = label_name,
             # categorical_features = feature_names,
             silent = True)
```

	Description	Value
0	session_id	1303
1	Target	label
2	Target Type	Binary
3	Label Encoded	0.0: 0, 1.0: 1
4	Original Data	(79, 12)
5	Missing Values	False
6	Numeric Features	10
7	Categorical Features	1
8	Ordinal Features	False
9	High Cardinality Features	False
10	High Cardinality Method	None
11	Transformed Train Set	(55, 11)
12	Transformed Test Set	(24, 11)
13	Shuffle Train-Test	True
14	Stratify Train-Test	False
15	Fold Generator	StratifiedKFold
16	Fold Number	10
17	CPU Jobs	-1
18	Use GPU	False
19	Log Experiment	False
20	Experiment Name	clf-default-name
21	USI	a02c
22	Imputation Type	simple
23	Iterative Imputation Iteration	None
24	Numeric Imputer	mean
25	Iterative Imputation Numeric Model	None
26	Categorical Imputer	constant
27	Iterative Imputation Categorical Model	None
28	Unknown Category Handling	least_frequent
29	Normalize	False
30	Normalize Method	None
31	Transformation	False
32	Transformation Method	None
33	PCA	False
34	PCA Method	None
35	PCA Components	None
36	Ignore Low Variance	False
37	Combine Rare Levels	False

	Description	Value
38	Rare Level Threshold	None
39	Numeric Binning	False
40	Remove Outliers	False
41	Outliers Threshold	None
42	Remove Multicollinearity	False
43	Multicollinearity Threshold	None
44	Remove Perfect Collinearity	True
45	Clustering	False
46	Clustering Iteration	None
47	Polynomial Features	False
48	Polynomial Degree	None
49	Trigonometry Features	False
50	Polynomial Threshold	None
51	Group Features	False
52	Feature Selection	False
53	Feature Selection Method	classic
54	Features Selection Threshold	None
55	Feature Interaction	False
56	Feature Ratio	False
57	Interaction Threshold	None
58	Fix Imbalance	False
59	Fix Imbalance Method	SMOTE

```
In [ ]: # 省時間全部fold切5分
lr = create_model('lr', fold = 25, max_iter=500)
# knn = create_model('knn', fold = 25)
# nb = create_model('nb', fold = 25)
# dt = create_model('dt', fold = 25)
# svm = create_model('svm', fold = 25)
# rbfsvm = create_model('rbfsvm', fold = 25)
# gpc = create_model('gpc', fold = 25)
# mlp = create_model('mlp', fold = 25)
# ridge = create_model('ridge', fold = 25)
rf = create_model('rf', fold = 25)
# qda = create_model('qda', fold = 25)
ada = create_model('ada', fold = 25)
# lda = create_model('lda', fold = 25)
# gbc = create_model('gbc', fold = 25)
et = create_model('et', fold = 25)
xgboost = create_model('xgboost', fold = 25)
lightgbm = create_model('lightgbm', fold = 25)
# catboost = create_model('catboost', fold = 25)

estimator_list = [lr,
                  # knn,
                  # nb,
                  # dt,
                  # svm,
                  # rbfsvm,
                  # gpc,
                  # mlp,
                  # ridge,
                  rf,
                  # qda,
                  ada,
                  # lda,
                  # gbc,
                  et,
                  xgboost,
                  lightgbm,
                  # catboost
                  ]

# 第二層用xgboost
stacker_all = stack_models(estimator_list = estimator_list, meta_model = xgboost)

pred = predict_model(stacker_all,data = df_validation)

accuracy_score(pred[label_name], pred[['Label']].astype(float))
```

	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Fold							
0	0.8333	0.6250	0.50	1.0000	0.6667	0.5714	0.6325
1	0.8333	0.8750	0.50	1.0000	0.6667	0.5714	0.6325
2	0.8333	1.0000	0.50	1.0000	0.6667	0.5714	0.6325
3	0.8333	1.0000	1.00	0.6667	0.8000	0.6667	0.7071
4	0.8333	0.6875	0.50	1.0000	0.6667	0.5714	0.6325
5	1.0000	1.0000	1.00	1.0000	1.0000	1.0000	1.0000
6	1.0000	1.0000	1.00	1.0000	1.0000	1.0000	1.0000
7	0.6000	0.6667	0.50	0.5000	0.5000	0.1667	0.1667
8	1.0000	1.0000	1.00	1.0000	1.0000	1.0000	1.0000
9	1.0000	1.0000	1.00	1.0000	1.0000	1.0000	1.0000
Mean	0.8767	0.8854	0.75	0.9167	0.7967	0.7119	0.7404
Std	0.1212	0.1529	0.25	0.1708	0.1792	0.2665	0.2543

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
0	Stacking Classifier	1.0	1.0	1.0	1.0	1.0	1.0	1.0

Out []: 1.0

```
In [ ]: y_true = pred['label']
y_pred = pred['Label'].astype(float)

print('classification_report() =\n', classification_report(y_true, list(y_pred), digits=5))

classification_report() =
precision    recall  f1-score   support
```



```
0.0    1.00000    1.00000    1.00000    6
1.0    1.00000    1.00000    1.00000    1

accuracy    1.00000    1.00000    1.00000    7
macro avg   1.00000    1.00000    1.00000    7
weighted avg 1.00000    1.00000    1.00000    7
```

Save stacker model

```
In [ ]: save_model(stacker_all, 'stacker_auc1')
        load_model('stacker_auc1')

Transformation Pipeline and Model Successfully Saved
Transformation Pipeline and Model Successfully Loaded

Out[ ]: Pipeline(memory=None,
                 steps=[('dtypes',
                        DataTypes_Auto_infer(categorical_features=[],
                                             display_types=False, features_todrop=[],
                                             id_columns=[],
                                             ml_usecase='classification',
                                             numerical_features=[], target='label',
                                             time_features=[])),
                        ('imputer',
                         Simple_Imputer(categorical_strategy='not_available',
                                          fill_value_categorical=None,
                                          fill_value_numerical=None,
                                          numeric_strat...

missing=nan,
monotone_constraints=(),
n_estimators=100,
n_jobs=-1,
num_parallel_tree=1,
objective='binary:logistic',
predictor='auto',
random_state=1303,
reg_alpha=0,
reg_lambda=1,
scale_pos_weight=1,
subsample=1,
tree_method='auto',
use_label_encoder=True,
validate_parameters=1,
verbosity=0),

n_jobs=-1, passthrough=True,
stack_method='auto', verbose=0)],

verbose=False)

In [ ]: dump(stacker_all, f'stacker_auc1.joblib')
        load(f'stacker_auc1.joblib')
```

Compress models/ folder

```
In [ ]: # %%sh
        # zip -r {folder_models}.zip {folder_models}

In [ ]: !echo "$(TZ=:Asia/Hong_Kong date +%Y%m%d.%H%Mm)"

In [ ]:
```

Test

```
In [ ]: def saveResult(df_result: np.array,
                      csv_name: str,
                      label_name: str = 'label',
                      folder = 'results'
                      ):
    """result.csv format
    | id | label |
    |---|-----|
    | xx | 0 or 1|
    """

    if not os.path.exists(folder):
        os.makedirs(folder)

    # np.savetxt(csv_name, result[['id', 'label']], delimiter=",")
    df_result[[label_name]].to_csv(f'{folder}/{csv_name}.csv', index=True)

In [ ]: ## load models
        path = 'stacker_auc1'
        # model = load_model('stacker_auc1')

        # pred_y = testModel(model, df_test[feature_names])
        df_result = predict_model(stacker_all, data = df_test)
        df_result = df_result.rename(columns={"Label": label_name})
        df_result[label_name] = df_result[label_name].astype(float).astype(int)

        display(df_result)

        csv_name = path[: path.find('.joblib')].replace(f'{folder_models}/', '')
        saveResult(df_result, csv_name, label_name, 'results')
        # break
```

	MO_HLADR_MFI_cellul	Neu_CD64MFI_cellul	CD3T_cellul	CD8T_cellul	CD4T_cellul	NK_cellul	CD19_cellul	CD45_cellul	Age	Sex_0M1F	Mono_CD64MFI_cellul	label	Score
id													
0	2843.0	156.0	1358.52	730.78	637.85	127.06	94.82	1588.62	45.0	1.0	3256.0	0	0.9675
1	437.0	137.0	509.43	268.05	243.07	390.86	98.24	1002.76	51.0	1.0	491.0	0	0.9731
2	826.0	82.0	1232.22	493.42	744.08	516.28	320.15	2200.58	32.0	0.0	1381.0	0	0.9741
3	861.0	50.0	1512.86	925.51	590.07	380.25	25.80	1929.10	50.0	0.0	1377.0	0	0.9811
4	1160.0	157.0	890.42	403.91	489.53	266.92	87.63	1251.52	43.0	0.0	1844.0	0	0.9853
5	867.0	85.0	1662.11	865.50	804.14	220.68	92.58	2063.11	44.0	1.0	986.0	0	0.9855
6	1330.0	114.0	1307.95	710.86	607.96	271.01	214.49	1855.05	31.0	0.0	2077.0	0	0.9488
7	494.0	48.0	1522.39	618.19	911.49	338.85	104.45	2013.05	36.0	1.0	1409.0	0	0.9911
8	2119.0	73.0	1219.66	732.14	468.48	71.54	83.08	1378.32	45.0	1.0	2403.0	0	0.8442
9	2052.0	39.0	1223.65	642.55	565.39	323.67	153.64	1711.88	36.0	0.0	1701.0	0	0.9879
10	4195.0	134.0	1513.79	776.94	752.60	110.28	142.50	1780.88	45.0	1.0	3160.0	0	0.9533
11	819.0	1000.0	794.33	391.72	407.35	312.61	113.20	1236.72	45.0	0.0	4086.0	1	0.9346

	MO_HLADR_MFI_cellul	Neu_CD64MFI_cellsul	CD3T_cellsul	CD8T_cellsul	CD4T_cellsul	NK_cellsul	CD19_cellsul	CD45_cellsul	Age	Sex_0M1F	Mono_CD64MFI_cellsul	label	Score
id													
12	685.0	60.0	470.04	114.98	321.32	174.86	31.49	679.19	62.0	0.0	737.0	0	0.9931
13	1330.0	114.0	1307.95	710.86	607.96	271.01	214.49	1855.05	31.0	0.0	2077.0	0	0.9488
14	82.0	30.0	693.25	426.73	272.51	956.78	120.53	1825.96	27.0	0.0	371.0	0	0.9923
15	781.0	43.0	1537.36	922.55	620.05	142.08	66.13	1771.11	36.0	0.0	930.0	0	0.9915
16	2140.0	125.0	2771.20	1738.55	1006.96	367.85	203.45	3355.86	27.0	1.0	2183.0	0	0.9295
17	1005.0	93.0	1061.65	379.12	683.06	99.71	76.25	1241.34	33.0	0.0	1269.0	0	0.9922
18	1202.0	39.0	716.76	272.92	447.73	89.55	121.08	931.68	54.0	1.0	643.0	0	0.9445
19	660.0	43.0	886.57	225.68	620.75	196.49	106.33	1202.41	54.0	0.0	814.0	0	0.9978
20	1289.0	272.0	583.00	206.00	373.00	18.00	59.00	660.00	29.0	0.0	2581.0	1	0.9863
21	2238.0	539.0	850.05	438.98	352.71	235.14	91.35	1182.46	44.0	0.0	4361.0	1	0.9385
22	127.0	81.0	435.00	259.00	166.00	57.00	60.00	558.00	54.0	1.0	856.0	0	0.9236
23	880.0	38.0	773.79	369.33	412.49	412.99	55.20	1257.03	34.0	0.0	800.0	0	0.9936
24	3232.0	72.0	2017.32	850.72	1176.28	440.86	501.91	2964.93	15.0	1.0	1517.0	0	0.9894
25	902.0	1141.0	258.01	168.63	99.18	45.46	8.83	314.25	50.0	1.0	3244.0	1	0.9860
26	922.0	167.0	2122.97	1443.00	689.42	87.11	122.22	2369.43	36.0	0.0	1995.0	1	0.6681
27	1520.0	67.0	829.56	293.00	511.00	403.13	144.62	1378.89	47.0	1.0	1700.0	0	0.9840
28	313.0	382.0	632.45	289.46	346.06	138.08	44.32	816.55	53.0	0.0	2656.0	1	0.9786
29	1240.0	59.0	1071.75	513.58	443.52	110.01	125.65	1311.46	50.0	1.0	1203.0	0	0.9904
30	1208.0	460.0	1492.00	422.00	1048.00	56.00	23.00	1571.00	55.0	1.0	1761.0	1	0.9939
31	601.0	1080.0	306.71	210.30	98.45	57.50	71.73	436.83	39.0	0.0	4413.0	1	0.9605
32	1814.0	143.0	627.33	268.55	376.05	155.43	46.63	833.27	52.0	0.0	1298.0	0	0.9225
33	232.0	615.0	352.11	221.62	131.79	74.31	170.87	601.47	41.0	0.0	1040.0	1	0.9507
34	1078.0	238.0	755.15	345.75	413.33	779.42	76.76	1648.73	52.0	0.0	2506.0	1	0.6919
35	1085.0	24.0	1970.56	1296.13	601.93	388.11	186.00	2549.06	34.0	0.0	639.0	0	0.9932
36	1851.0	95.0	1498.79	983.18	525.76	147.51	50.75	1701.79	45.0	1.0	2347.0	0	0.9831
37	402.0	378.0	544.54	227.96	327.85	283.54	36.11	898.00	39.0	0.0	1849.0	1	0.9502
38	909.0	574.0	326.77	244.91	80.39	42.97	16.01	386.90	47.0	0.0	3969.0	1	0.9860
39	1296.0	148.0	1328.00	642.00	670.00	666.11	266.87	2267.31	51.0	0.0	2061.0	0	0.9268
40	1784.0	474.0	337.75	133.11	204.64	25.50	16.00	379.25	42.0	0.0	6788.0	1	0.9782
41	676.0	64.0	622.25	345.00	221.00	50.60	56.04	731.13	29.0	0.0	1660.0	0	0.8800
42	629.0	76.0	427.06	240.16	138.57	83.83	2.96	525.20	42.0	0.0	1391.0	0	0.9378
43	721.0	117.0	983.00	449.00	539.00	141.00	76.00	1200.00	37.0	0.0	2771.0	0	0.9654
44	1733.0	134.0	1443.62	770.62	613.22	85.40	227.47	1760.90	42.0	1.0	1501.0	0	0.7609
45	1546.0	184.0	832.08	621.01	210.58	141.36	63.86	1043.63	51.0	0.0	2281.0	0	0.9360
46	838.0	226.0	1025.32	728.71	296.61	187.36	135.49	1358.55	38.0	1.0	1662.0	0	0.9272
47	740.0	187.0	530.77	154.69	377.34	40.07	45.14	618.76	31.0	0.0	2617.0	1	0.9823
48	494.0	61.0	1568.54	937.00	601.00	246.48	165.92	1983.61	38.0	0.0	1923.0	0	0.9939
49	1700.0	71.0	1106.54	503.71	547.54	363.45	176.67	1650.03	48.0	0.0	1617.0	0	0.9817
50	1175.0	60.0	847.11	362.52	486.42	512.66	116.57	1486.10	47.0	0.0	2013.0	0	0.9908
51	1956.0	152.0	1786.48	783.82	952.42	194.75	198.19	2188.36	40.0	1.0	2251.0	0	0.9606
52	1122.0	66.0	1955.03	1080.16	879.77	111.60	100.20	2201.04	30.0	0.0	1468.0	0	0.9926
53	2492.0	60.0	2278.56	937.47	1225.68	252.10	304.97	2864.00	21.0	1.0	924.0	0	0.9860
54	708.0	248.0	1304.86	433.61	789.19	257.43	122.28	1691.81	37.0	1.0	1669.0	0	0.9456
55	560.0	74.0	727.01	227.25	502.48	159.80	106.38	1011.29	54.0	0.0	1235.0	0	0.9961
56	1948.0	264.0	1836.00	838.96	819.69	184.56	105.53	1943.45	35.0	1.0	2126.0	0	0.8833
57	1010.0	90.0	1061.65	379.12	683.06	99.71	76.25	1241.34	34.0	0.0	1319.0	0	0.9922
58	674.0	380.0	393.49	256.02	137.27	17.72	63.81	479.55	40.0	0.0	3095.0	1	0.9705

In []:

df_test

Out[]:

MO_HLADR_MFI_cellul

Neu_CD64MFI_cellsul

CD3T_cellsul

CD8T_cellsul

CD4T_cellsul

NK_cellsul

CD19_cellsul

CD45_cellsul

Age

Sex_0M1F

Mono_CD64MFI_cellsul

id													
0.0	2843.0	156.0	1358.52	730.78	637.85	127.06	94.82	1588.62	45.0	1.0	3256.0		
1.0	437.0	137.0	509.43	268.05	243.07	390.86	98.24	1002.76	51.0	0.0	491.0		
2.0	826.0	82.0	1232.22	493.42	744.08	516.28	320.15	2200.58	32.0	1.0	1381.0		
3.0	861.0	50.0	1512.86	925.51	590.07	380.25	25.80	1929.10	50.0	0.0	1377.0		
4.0	1160.0	157.0	890.42	403.91	489.53	266.92	87.63	1251.52	43.0	0.0	1844.0		
5.0	867.0	85.0	1662.11	865.50	804.14	220.68	92.58	2063.11	44.0	1.0	986.0		
6.0	1330.0	114.0	1307.95	710.86	607.96	271.01	214.49	1855.05	31.0	0.0	2077.0		
7.0	494.0	48.0	1522.39	618.19	911.49	338.85	104.45	2013.05	36.0	1.0	1409.0		
8.0	2119.0	73.0	1219.66	732.14	468.48	71.54	83.08	1378.32	45.0	1.0	2403.0		
9.0	2052.0	39.0	1223.65	642.55	565.39	323.67	153.64	1711.88	36.0	0.0	1701.0		
10.0	4195.0	134.0	1513.79	776.94	752.60	110.28	142.50	1780.88	45.0	1.0	3160.0		
11.0	819.0	1000.0	794.33	391.72	407.35	312.61	113.20	1236.72	45.0	0.0	4086.0		
12.0	685.0	60.0	470.04	114.98	321.32	174.86	31.49	679.19	62.0	0.0	737.0		
13.0	1330.0	114.0	1307.95	710.86	607.96	271.01	214.49	1855.05	31.0	0.0	2077.0		
14.0	82.0	30.0	693.25	426.73	272.51	956.78	120.53	1825.96	27.0	0.0	371.0		
15.0	781.0	43.0	1537.36	922.55	620.05	142.08	66.13	1771.11	36.0	0.0	930.0		
16.0	2140.0	125.0	2771.20	1738.55	1006.96	367.85	203.45	3355.86	27.0	1.0	2183.0		
17.0	1005.0	93.0	1061.65	379.12	683.06	99.71	76.25	1241.34	33.0	0.0	1269.0		
18.0	1202.0	39.0	716.76	272.92	447.73	89.55	121.08	931.68	54.0	1.0	643.0		
19.0	660.0	43.0	886.57	225.68	620.75	196.49	106.33	1202.41	54.0	0.0	814.0		
20.0	1289.0	272.0	583.00	206.00	373.00	18.00	59.00	660.00	29.0	0.0	2581.0		
21.0	2238.0	539.0	850.05	438.98	352.71	235.14	91.35	1182.46	44.0	0.0	4361.0		
22.0	127.0	81.0	435.00	259.00	166.00	57.00	60.00	558.00	54.0	1.0	856.0		

	MO_HLADR_MFI_cellsul	Neu_CD64MFI_cellsul	CD3T_cellsul	CD8T_cellsul	CD4T_cellsul	NK_cellsul	CD19_cellsul	CD45_cellsul	Age	Sex_0M1F	Mono_CD64MFI_cellsul
id											
23.0	880.0	38.0	773.79	369.33	412.49	412.99	55.20	1257.03	34.0	0.0	800.0
24.0	3232.0	72.0	2017.32	850.72	1176.28	440.86	501.91	2964.93	15.0	1.0	1517.0
25.0	902.0	1141.0	258.01	168.63	99.18	45.46	8.83	314.25	50.0	1.0	3244.0
26.0	922.0	167.0	2122.97	1443.00	689.42	87.11	122.22	2369.43	36.0	0.0	1995.0
27.0	1520.0	67.0	829.56	293.00	511.00	403.13	144.62	1378.89	47.0	1.0	1700.0
28.0	313.0	382.0	632.45	289.46	346.06	138.08	44.32	816.55	53.0	0.0	2656.0
29.0	1240.0	59.0	1071.75	513.58	443.52	110.01	125.65	1311.46	50.0	1.0	1203.0
30.0	1208.0	460.0	1492.00	422.00	1048.00	56.00	23.00	1571.00	55.0	1.0	1761.0
31.0	601.0	1080.0	306.71	210.30	98.45	57.50	71.73	436.83	39.0	0.0	4413.0
32.0	1814.0	143.0	627.33	268.55	376.05	155.43	46.63	833.27	52.0	0.0	1298.0
33.0	232.0	615.0	352.11	221.62	131.79	74.31	170.87	601.47	41.0	0.0	1040.0
34.0	1078.0	238.0	755.15	345.75	413.33	779.42	76.76	1648.73	52.0	0.0	2506.0
35.0	1085.0	24.0	1970.56	1296.13	601.93	388.11	186.00	2549.06	34.0	0.0	639.0
36.0	1851.0	95.0	1498.79	983.18	525.76	147.51	50.75	1701.79	45.0	1.0	2347.0
37.0	402.0	378.0	544.54	227.96	327.85	283.54	36.11	898.00	39.0	0.0	1849.0
38.0	909.0	574.0	326.77	244.91	80.39	42.97	16.01	386.90	47.0	0.0	3969.0
39.0	1296.0	148.0	1328.00	642.00	670.00	666.11	266.87	2267.31	51.0	0.0	2061.0
40.0	1784.0	474.0	337.75	133.11	204.64	25.50	16.00	379.25	42.0	0.0	6788.0
41.0	676.0	64.0	622.25	345.00	221.00	50.60	56.04	731.13	29.0	0.0	1660.0
42.0	629.0	76.0	427.06	240.16	138.57	83.83	2.96	525.20	42.0	0.0	1391.0
43.0	721.0	117.0	983.00	449.00	539.00	141.00	76.00	1200.00	37.0	0.0	2771.0
44.0	1733.0	134.0	1443.62	770.62	613.22	85.40	227.47	1760.90	42.0	1.0	1501.0
45.0	1546.0	184.0	832.08	621.01	210.58	141.36	63.86	1043.63	51.0	0.0	2281.0
46.0	838.0	226.0	1025.32	728.71	296.61	187.36	135.49	1358.55	38.0	1.0	1662.0
47.0	740.0	187.0	530.77	154.69	377.34	40.07	45.14	618.76	31.0	0.0	2617.0
48.0	494.0	61.0	1568.54	937.00	601.00	246.48	165.92	1983.61	38.0	0.0	1923.0
49.0	1700.0	71.0	1106.54	503.71	547.54	363.45	176.67	1650.03	48.0	0.0	1617.0
50.0	1175.0	60.0	847.11	362.52	486.42	512.66	116.57	1486.10	47.0	0.0	2013.0
51.0	1956.0	152.0	1786.48	783.82	952.42	194.75	198.19	2188.36	40.0	1.0	2251.0
52.0	1122.0	66.0	1955.03	1080.16	879.77	111.60	100.20	2201.04	30.0	0.0	1468.0
53.0	2492.0	60.0	2278.56	937.47	1225.68	252.10	304.97	2864.00	21.0	1.0	924.0
54.0	708.0	248.0	1304.86	433.61	789.19	257.43	122.28	1691.81	37.0	1.0	1669.0
55.0	560.0	74.0	727.01	227.25	502.48	159.80	106.38	1011.29	54.0	0.0	1235.0
56.0	1948.0	264.0	1836.00	838.96	819.69	184.56	105.53	1943.45	35.0	1.0	2126.0
57.0	1010.0	90.0	1061.65	379.12	683.06	99.71	76.25	1241.34	34.0	0.0	1319.0
58.0	674.0	380.0	393.49	256.02	137.27	17.72	63.81	479.55	40.0	0.0	3095.0